

Retail Sales and Profitability Performance Analysis Report

SUPERSTORE

Power BI Final Project Report

Dataset: Fictional Superstore Retail Transaction Dataset

Analysis period: 2023-2026

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1. Executive Summary

This report analyses a fictional Superstore retail dataset using Power BI. The analysis focuses on sales, profit, profit margin, product categories, customer segments, regions, discount levels, shipping modes, and time trends. The main business problem is that high sales do not always lead to high profit. Therefore, the report evaluates both revenue and profitability measures to identify business areas that are performing well and areas that reduce overall profitability.

- Overall sales and profit increased from 2023 to 2026, but profit margin did not increase consistently.
- Technology and Office Supplies are the strongest categories by profit, while Furniture has weak profitability and the largest losses.
- The United States dominates regional sales, especially California and New York, but high-sales states such as Texas may still show weak profit margins.
- Consumer is the largest customer segment, followed by Corporate and Home Office.
- Medium and high discounts create profitability risks because they increase loss-making orders and reduce profit per order.
- Standard Class appears to be the most stable shipping option, while faster shipping modes show more variable profit per order.
- Quarter 4 (Q4) is the strongest seasonal period, while Quarter 1 (Q1) is generally weaker.

The report recommends controlling high discounts, reviewing Furniture-related losses, prioritising profitable products and regions, preparing operational resources before Q4, and using profit margin together with sales when making business decisions.

2. Introduction

2.1. Business Context

The dataset represents a fictional retail Superstore that sells products across different customer segments, regions, states, cities, product categories, and sub-categories. Each transaction includes sales-related information such as order date, ship date, sales amount, quantity, discount, profit, customer segment, product details, and shipping mode.

This project uses Power BI to understand how the Superstore performs in terms of sales, profit, discount impact, customer segments, product performance, regional performance, shipping mode, and time-based trends.

2.2. Objectives

The main objective of this project is to analyse the Superstore transactional retail dataset using Power BI to identify key patterns in sales and profitability performance. The analysis aims to create interactive dashboards that help management understand profitable and loss-making areas of the business.

- Conduct exploratory data analysis on retail transactions;
- Create interactive Power BI dashboards;
- Identify profitable and loss-making regions, categories, sub-categories, and products;
- Analyse how discounts and shipping modes affect profit;
- Understand customer purchasing behaviour across segments;
- Examine sales and profit trends over time;
- Provide practical recommendations to improve retail performance.

2.3. Business Problem

The Superstore generates sales across multiple regions, customer segments, and product categories. However, high sales do not always mean strong profit. Some products, regions, or customer groups may generate large sales but low or negative profit due to high discounts, inefficient shipping modes, weak product performance, or regional differences.

The business needs to identify which areas drive profit and which areas reduce profitability so that management can make better decisions about pricing, discount strategy, product focus, regional targeting, customer segmentation, and shipping operations.

2.4. Business Questions

- (1) Which product categories and sub-categories are most profitable or loss-making, and in which regions is this problem most common?
- (2) Which regions and states generate the highest sales and profit, and do high-sales regions also have strong profit margins?
- (3) How do discounts affect profit performance, and are high discounts reducing profit without significantly increasing quantity sold?
- (4) Which customer segment, Consumer, Corporate, or Home Office, contributes the most to total sales and long-term profitability?
- (5) Does ship mode affect profit per order?
- (6) Which products have high sales but low or negative profit?

(7) Based on order date, are there specific months, quarters, or seasons where the Superstore experiences significant peaks or dips in sales and profit?

3. Data Preparation and Modelling

3.1. Dataset Tables and Data Quality Check

Table 1: Overview of Source Dataset Tables

Table	Role	Meaning
Orders	Fact/source table	Main transaction table containing sales, profit, discount, customer, product, region, and shipping fields.
People	Supporting table	Contains regional manager/person information.
Returns	Supporting lookup table	Contains returned order records.

Data quality checking was completed in Power Query using column quality, column distribution, and column profile. No major missing value, error value, or data type issue was found in the main analytical fields. Duplicate checks confirmed that Row ID in the Orders table was unique. Repeated Order ID values were retained because one order can contain multiple product lines.

Table 2: Data Quality Assessment Summary

Check	Result	Interpretation
Missing values	No major missing fields in main analytical columns	The dataset is suitable for analysis.
Errors	No major error values found	No major data cleaning issue.
Data types	Corrected/checked before modelling	Fields can be used correctly in measures and visuals.

3.2. Analytical Columns Created

Table 3: Calculated Columns Used for Profit, Discount and Shipping Analysis

Column	Logic / Definition	Purpose
Profit Status	Profit < 0 = Loss; otherwise Profit	Classifies each transaction as profitable or loss-making.
Discount Level	0 = No; >0 to <=0.2 = Low; >0.2 to <=0.5 = Medium; >0.5 = High	Groups discounts into readable categories for analysis.
Shipping Days	Duration.Days([Ship Date] - [Order Date])	Measures the number of days between order and shipment.

3.3. Star Schema Data Model

The original table Orders table was transformed into a star schema. A new FactSales table was created as the central transaction-level table, while dimension tables were created to support filtering and grouping by product, customer, location, shipping mode, discount level, and date.

Table 4: Power BI Star Schema Model Design

Table	Type	Key fields / content	Purpose
FactSales	Fact table	Sales, Profit, Quantity, Discount, Order ID, Customer ID, Product ID, Order Date, Ship Date, Shipping Days	Stores transaction-level numerical data.
Dim_Product	Dimension	Product ID, Product Name, Category, Sub-Category	Supports category, sub-category, and product analysis.
Dim_Customers	Dimension	Customer ID, Customer Name, Segment	Supports Consumer, Corporate, and Home Office analysis.
Dim_Location	Dimension	Location Key, Country, Region, State, City, Postal Code	Supports geographical analysis.
Dim_ShipMode	Dimension	Ship Mode	Supports shipping efficiency analysis.
Dim_DiscountLevel	Dimension	Discount Level, Discount Sort	Supports discount impact analysis.
Dim_Date	Dimension	Order Date, Year, Quarter, Month, Year-Month	Supports trend and seasonality analysis.

Relationships were created as one-to-many relationships from each dimension table to FactSales. Cross-filter direction was set to single and active relationships were used. This model design allows dimension fields to filter transaction-level measures clearly and avoids unnecessary complexity.

3.4. Key DAX Measures

Table 5: Core DAX Measures for Sales and Profitability Analysis

Measure	DAX / Definition	Purpose
Total Sales	SUM(FactSales[Sales])	Total revenue generated.
Total Profit	SUM(FactSales[Profit])	Total profitability.
Total Quantity	SUM(FactSales[Quantity])	Total units sold.
Total Orders	DISTINCTCOUNT(FactSales[Order ID])	Number of orders.
Total Customers	DISTINCTCOUNT(FactSales[Customer ID])	Number of customers.
Profit Margin %	DIVIDE([Total Profit], [Total Sales])	Profitability efficiency.
Average Discount	AVERAGE(FactSales[Discount])	Average discount rate.
Profit per Order	DIVIDE([Total Profit], [Total Orders])	Average profit generated per order.
Average Shipping Days	AVERAGE(FactSales[Shipping Days])	Average delivery duration.
Loss Orders	CALCULATE(DISTINCTCOUNT(FactSales[Order ID]), FactSales[Profit] < 0)	Number of loss-making orders.
Loss Order %	DIVIDE([Loss Orders], [Total Orders])	Proportion of orders that generated losses.
Loss Amount	CALCULATE([Total Profit], FactSales[Profit] < 0)	Total negative profit amount.
Profit Amount	CALCULATE([Total Profit], FactSales[Profit] > 0)	Total positive profit amount.

4. Dashboard Visualisation Design

There are four interactive dashboard was developed in Power BI. The dashboard follows a progressive flow from overview to detailed investigation: Executive Overview, Product Profitability Analysis, Regional and Customer Segment Analysis, and Discount, Shipping and Trend Analysis. A navigation page and Home buttons were used to improve user experience.

Table 6: Dashboard Structure and Analytical Purpose

Dashboard page	Main purpose	Main visuals
Executive Overview	Provide high-level business performance summary.	KPI cards, sales and profit trend, category comparison, profit margin by category, least profitable sub-categories.
Product Profitability Analysis	Identify profitable and loss-making categories, sub-categories, and products.	Loss/profit KPI cards, profit by category/sub-category, profitability risk scatter, lowest-profit product table, loss orders by sub-category.
Regional & Customer Segment Analysis	Analyse geographical performance and customer segment contribution.	Profit per order KPI, sales and margin by states, lowest-profit cities, state profitability by segment, segment sales by top states.
Discount, Shipping & Trend Analysis	Analyse discount, shipping and time-based operational factors.	Sales/profit trend, order volume and profit per order, margin and loss rate trend, shipping time vs profit per order.

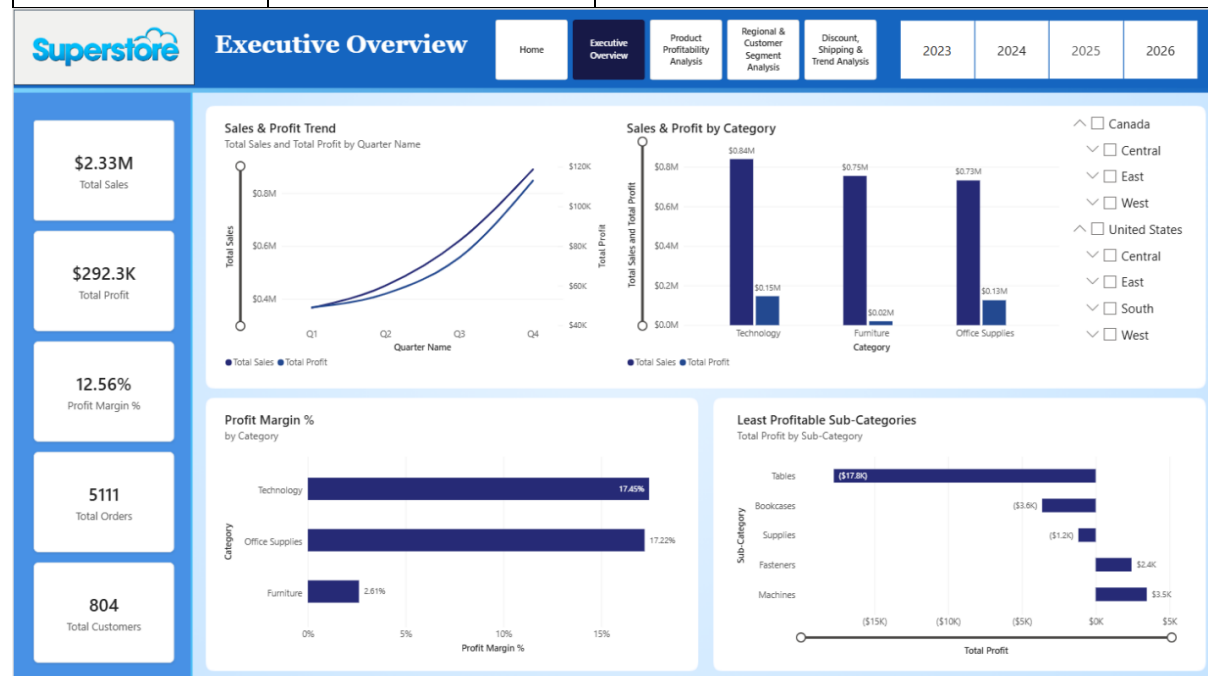


Figure 1: Executive Overview dashboard screenshot

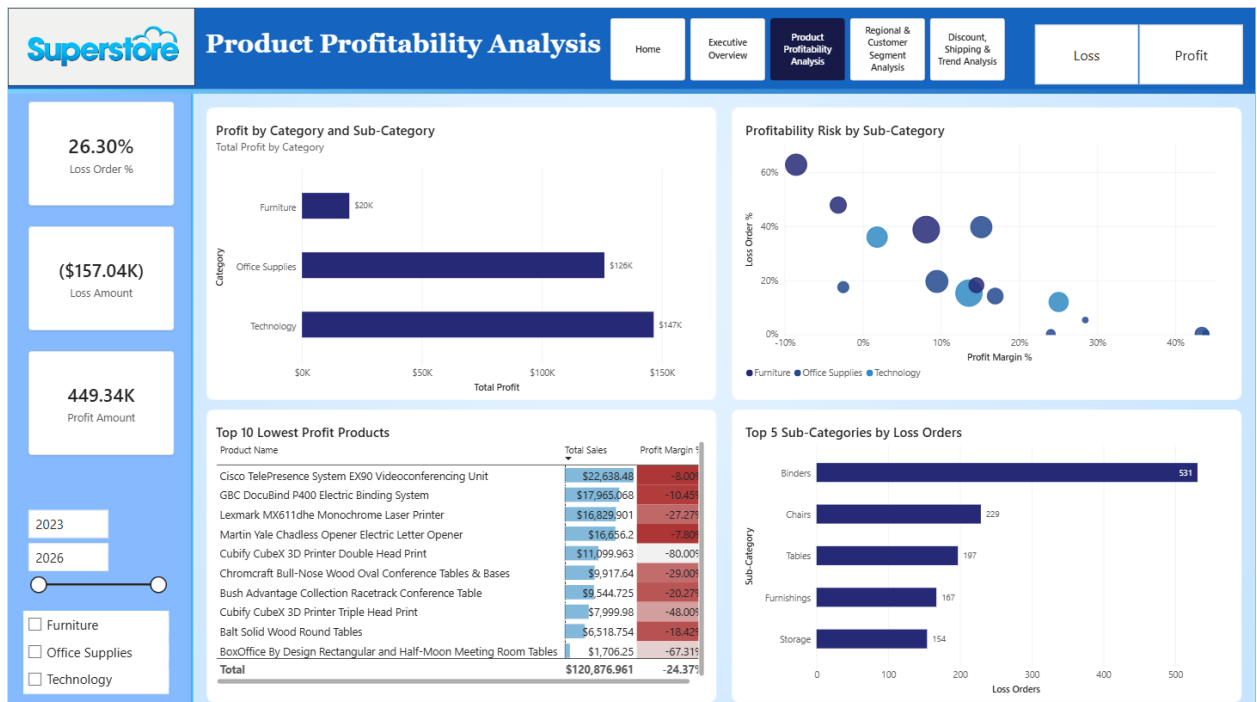


Figure 2: Product Profitability Analysis dashboard screenshot



Figure 3: Regional & Customer Segment Analysis dashboard screenshot



Figure 4: Discount, Shipping & Trend Analysis dashboard screenshot

Table 7: Dashboard Visuals and Supporting Measures

Visual / Table	Fields / Measures used	How it supports analysis
Cards	Total Sales, Total Profit, Profit Margin %, Total Orders, Total Customers	Summarises overall business performance.
Sales & Profit Trend	Quarter Name on X-axis; Total Sales and Total Profit as values	Shows seasonal sales and profit movement.
Category comparison chart	Category; Total Sales; Total Profit	Compares revenue and profit by category.
Profit Margin by Category	Category; Profit Margin %	Shows profitability efficiency by category.
Loss-making product table	Product Name; Total Sales; Total Profit; Profit Margin %; visual filter by Loss	Identifies high-sales but low/negative-profit products.
Top states analysis	State; Total Sales; Profit Margin %; optional Total Profit	Compares regional sales and margin performance.
Segment sales by state	State; Segment; Total Sales	Shows which customer segment contributes most in each top state.
Discount impact visuals	Discount Level; Total Sales; Total Orders; Profit per Order; Profit Margin %; Loss Order %	Evaluates whether discounts improve sales or reduce profit.
Shipping efficiency scatter	Average Shipping Days; Profit per Order; Total Orders; Ship Mode slicer	Compares shipping duration and order profitability.

5. Data Analysis and Findings

5.1 Executive Overview

The Executive Overview shows that the Superstore is profitable overall, with total sales of approximately \$2.33M and total profit of \$292.3K across 2023-2026. Sales and profit generally increased over time, but profit margin did not improve consistently.

Table 8: Annual Sales, Profit and Margin Performance

Year	Total Sales	Total Profit	Profit Margin	Main insight
2023	\$494.04K	\$51.68K	10.46%	Lowest margin; weaker profit efficiency.
2024	\$477.99K	\$62.02K	13.11%	Sales slightly lower than 2023 but profit improved.
2025	\$613.90K	\$82.67K	13.46%	Strong growth and best margin.
2026	\$745.57K	\$95.93K	12.87%	Highest sales and profit, but margin dropped slightly.
Average	\$2.33M	\$292.30K	12.56%	Business is profitable overall.

A key insight is that 2024 had lower sales than 2023 but higher profit and a higher profit margin. This proves that sales volume alone does not fully explain business performance. Profitability also depends on product mix, discounting, and cost-related factors.

5.2 Profitability Analysis

Business question: Which categories and sub-categories are most profitable or loss-making?

Table 9: Category Profitability Comparison

Category	Positive profit view	Loss view	Interpretation
Technology	Around \$185K profit	Around - \$39K loss	Strongest profit-generating category overall.
Office Supplies	Around \$183K profit	Around - \$57K loss	Strong contributor but still has loss-making sub-categories.
Furniture	Around \$81K profit	Around - \$62K loss	Weakest category; largest losses and low net profitability.

Table 10: Table 10: Major Loss-Making Sub-Categories

Loss-making sub-category	Loss orders	Interpretation
Binders	531	Most frequent source of loss-making orders.
Chairs	229	Major Furniture-related loss contributor.
Tables	197	High-value loss driver and important Furniture issue.
Furnishings	167	Another Furniture-related loss area.
Storage	154	Repeated loss-making orders within Office Supplies.

The most profitable categories are Technology and Office Supplies, with positive profits of around \$185K and \$183K respectively. Technology is the strongest category because it generates the highest profit while having the lowest loss amount, around -\$39K. Office Supplies is also a strong contributor, but it has a larger loss amount of around -\$57K, mainly because Binders have the highest number of loss-making orders at 531.

In contrast, Furniture is the weakest category. It only generates around \$81K positive profit, but has the largest loss amount, around -\$62K. This shows that Furniture has a much weaker profitability position than the other categories. The main loss-making sub-categories are Chairs, Tables, and Furnishings, with 229, 197, and 167 loss orders respectively.

A key insight is that loss problems are not only caused by one category. Office Supplies has the most frequent loss-making sub-category through Binders, but Furniture creates the biggest overall concern because several of its sub-categories contribute to losses at the same time. In particular, Tables and Chairs should be reviewed carefully, because they may involve higher product costs, higher shipping costs, discounting issues, or lower profit margins.

Overall, the business should continue supporting Technology and Office Supplies because they are the main profit drivers. However, it should also investigate the pricing, discounting, and shipping strategy for Furniture, especially Tables and Chairs. For Binders, the business should review why so many orders are loss-making, such as whether discounts are too high or whether the selling price is too low. Reducing losses in these sub-categories could significantly improve overall profitability.

5.3 Regional Performance

Business question: Which regions and states generate the highest sales and profit, and do high-sales regions also have strong profit margins?

Table 11: Country and Regional Performance Summary

Area	Finding	Meaning
United States	Dominates total sales and regional performance.	Most high-sales states are from the US.
Canada	Much smaller sales volume.	Canada has less impact on the combined dashboard.
Main high-sales states	California, New York, Texas, Washington, Pennsylvania.	These states are key revenue drivers.
Profitability issue	High sales do not always mean high profit margin.	Some states generate sales but have weak profitability.

Table 12: State-Level Sales and Profit Margin Comparison

State / location	Sales performance	Profit margin performance	Interpretation
California	Very high sales	Generally positive margin	Strong regional performer.
New York	Very high sales	Generally positive margin	Strong regional performer.
Texas	High sales	Weak or negative margin in some views	High revenue but profitability risk.
Ontario / Alberta / Quebec	Important in Canada	Smaller volume than US states	Useful for country-level analysis but less dominant overall.

The regional analysis shows that the United States is the main driver of total sales and profit, while Canada has a much smaller impact on the combined dashboard. The main high-sales states are California, New York, Texas, Washington, and Pennsylvania.

California and New York are the strongest performers because they generate very high sales and generally maintain positive profit margins. This means they are not only high-revenue states, but also profitable markets.

However, high sales do not always lead to strong profit margins. Texas is a key example because it has high sales but shows weak or negative profit margin in some views. This suggests that Texas may have issues with discounting, product mix, pricing, or shipping costs.

The lowest-profit city visuals also show repeated problem areas such as Houston, Philadelphia, Chicago, Dallas, Louisville, Jacksonville, Phoenix, and Aurora. These cities should be reviewed to identify what is reducing profitability.

Overall, the key insight is that California and New York should be maintained as strong markets, while Texas and low-profit cities need further investigation. Regional decisions should consider sales, profit, and profit margin together, rather than relying only on sales volume.

5.4 Discount Impact

Business question: How do discounts affect profit performance, and are high discounts reducing profit without significantly increasing quantity sold?

Table 13: Discount Impact on Sales and Profitability

Discount level	Sales / order pattern	Profit pattern	Loss risk	Interpretation
No discount	Strong sales and order volume, especially in later quarters.	Profit remains positive.	Low	Best profitability condition.
Low discount	Sales and orders increase, especially toward Q4.	Profit remains mostly positive.	Medium	Acceptable if controlled.
Medium discount	Generates some sales and orders.	Profit becomes weak or negative.	High	Risky discount level.
High discount	Does not clearly generate enough additional sales/orders.	Profit becomes negative.	Very high	Most damaging to profitability.

The discount analysis shows that discounts can increase sales and order volume, but higher discounts do not always improve profitability. Orders with no discount perform best because they still generate strong sales and order volume while keeping profit positive. This shows that the business does not always need discounts to attract customers, and no-discount orders are the healthiest condition for profit performance.

Low discounts can still be useful because they support sales and orders, especially toward later quarters, while profit remains mostly positive. However, they should be used in a controlled way because even small discounts reduce the profit earned per order.

The main issue appears with medium and high discounts. Medium discounts may generate some sales, but profit becomes weak or negative, which means the extra sales are not enough to cover the reduced margin. High discounts are the most damaging, because they do not clearly increase sales or quantity sold enough to justify the profit loss. Instead, they create negative profit and a higher risk of loss-making orders.

The key insight is that discounts should not be evaluated by sales volume only. A high-discount order may look successful because it creates revenue, but if the profit margin is negative, the business is losing money from that sale. Therefore, the business should continue using no discount and low discount strategies where they remain profitable, but carefully review or limit medium and high discounts, especially for weak-margin products, categories, or regions.

5.5 Customer Segmentation

Business question: Which customer segment contributes the most to total sales and long-term profitability?

Table 14: Customer Segment Sales and Profit Contribution

Segment	Sales contribution	Profitability indication	Interpretation
Consumer	Highest contribution in most top states.	Strong overall contribution.	Main customer segment and biggest revenue driver.
Corporate	Second strongest contribution.	Stable contribution in major states.	Important secondary segment with long-term value.
Home Office	Smallest contribution.	Lower total contribution.	Smaller segment, but still useful in selected states.

Table 15: Segment Performance Across Key States

State	Main segment finding
California	Highest sales state overall; all three segments contribute, but Consumer is the largest.
New York	Second major sales state with strong Consumer and Corporate contribution.
Texas	High sales but weaker profitability in some views.
Washington / Pennsylvania / Florida	Moderate sales contribution across segments.

The customer segment analysis shows that Consumer is the most important segment because it contributes the highest total sales in most key states, especially California and New York. This means Consumer is the main driver of overall revenue and has the biggest influence on business performance. Since Consumer appears strongly across high-sales states, this segment should remain the main focus for marketing, sales campaigns, and customer retention strategies.

Corporate is the second strongest segment, but its value is not only in sales volume. It provides a more stable contribution across major states, which suggests that Corporate customers may support long-term profitability through repeated or higher-value purchases. Although Corporate does not generate as much revenue as Consumer, it is still important because it can provide steady demand and reduce the business's dependence on Consumer customers alone.

Home Office has the smallest overall contribution, so it has less impact on total sales and profit compared with the other two segments. However, this segment should not be removed from consideration completely. Its smaller size may mean fewer orders, but some Home Office customers may still be profitable in certain states or product groups. Therefore, the business should review Home Office at a more detailed level before deciding whether to reduce focus on this segment.

A key insight is that the highest-sales segment is not automatically the most profitable in every location. For example, Texas has high sales but weaker profitability in some views, which means some segment-state combinations may create revenue without strong margins. This shows that customer segment performance should be analysed together with profit margin, total profit, and profit per order, not only total sales.

Overall, Consumer is the strongest segment and should remain the main target, while Corporate should be maintained as an important long-term segment. Home Office is smaller but may still have selective value. The business should focus on growing Consumer and Corporate in profitable states, while reviewing weaker areas such as Texas to understand whether discounts, product mix, or shipping costs are reducing profitability.

5.6 Shipping Efficiency

Business question: Does ship mode affect profit per order?

Table 16: Shipping Mode Efficiency and Profitability

Ship mode	Expected pattern from dashboard	Business interpretation
Standard Class	Higher order volume and longer shipping time.	Likely the most stable shipping option.
Second Class	Medium shipping time and moderate order volume.	May provide a balance between speed and profitability.
First Class	Faster shipping and lower order volume.	Profit per order may be less stable.
Same Day	Fastest shipping and lowest order volume.	More volatile; should be monitored carefully.

The dashboard suggests that ship mode does affect profit per order, but the impact is not simply about whether the delivery is faster or slower. Faster shipping does not automatically

lead to better profitability. Instead, profit per order depends on a combination of shipping cost, order volume, product margin, and customer demand.

Standard Class appears to be the most stable shipping option. It usually has the highest order volume and longer shipping time, but this does not make it less valuable. Because many customers use this option, it can support consistent sales and more stable profit per order. This suggests that Standard Class is likely suitable for regular orders where speed is not the main priority.

Second Class provides a middle position between speed and stability. It has moderate shipping time and order volume, so it may offer a balance between customer convenience and profitability. This shipping mode can be useful when customers want faster delivery than Standard Class, but the business still needs to control shipping-related costs.

First Class and Same Day require more careful review. Although they are faster, they have lower order volume and may show more unstable profit per order. This means faster delivery may increase service quality, but it can also reduce profitability if the shipping cost is too high or if the order value is not large enough to cover the cost.

The key insight is that speed should not be the only factor when evaluating ship mode performance. A faster shipping option may look attractive from a customer-service perspective, but it may not be the best option financially. The business should compare profit per order, order volume, and shipping time together before deciding which ship mode to promote.

Overall, Standard Class seems to be the safest and most stable shipping option, while Same Day and First Class should be monitored more carefully. The business should investigate whether fast shipping is being used for low-value or low-margin orders, because this could reduce profit per order.

5.7 Product Performance

Business question: Which products have high sales but low or negative profit?

Table 17: Category-level product performance

Category / evidence	Dashboard number	Interpretation
Technology	Profit amount: \$185.12K; loss amount: -\$38.58K; net category profit: about \$147K	Highest profit contribution and strongest net category profit.
Office Supplies	Profit amount: \$182.81K; loss amount: -\$56.78K; net category profit: about \$126K	Second strongest category, but losses are higher than Technology.
Furniture	Profit amount: \$81.41K; loss amount: -\$61.68K; net category profit: about \$19.7K	Lowest net profit and largest loss amount. This is the weakest category.
All categories	Total profit amount: \$449.34K; total loss amount: -\$157.04K; loss order %: 26.30%	About one quarter of orders are loss-making.

At category level, Technology is the strongest product category. It has the highest profit amount, around \$185.12K, and a much lower loss amount of around -\$38.58K. This gives Technology the highest net profit, about \$147K. This means Technology is not only a high-sales category, but also the most profitable overall.

Office Supplies is the second strongest category, with around \$182.81K profit and about \$126K net profit. However, its loss amount is higher than Technology, at around -\$56.78K. This shows that Office Supplies still performs well overall, but some products inside this category are creating repeated losses. The clearest issue is Binders, which has 531 loss-making orders, the highest among all sub-categories.

Furniture is the weakest product category. Although it still creates around \$81.41K positive profit, it also has the largest loss amount, around -\$61.68K. As a result, its net profit is only about \$19.7K, much lower than Technology and Office Supplies. This means Furniture has serious profitability problems. The main risky sub-categories are Chairs, Tables, and Furnishings, which together show repeated loss-making orders.

Table 18: Loss-making sub-category evidence

Sub-category	Loss orders	Why it is important
Binders	531	Highest loss-order count; most frequent repeated loss source.
Chairs	229	Second-highest loss-order count; major Furniture-related issue.
Tables	197	Third-highest loss-order count; linked to weak Furniture performance.
Furnishings	167	Another Furniture-related source of repeated losses.
Storage	154	Lowest within the top five, but still a recurring loss source.

Table 19: High-sales products with low or negative profit margin

Product example from lowest-profit table	Total sales	Profit margin
Cisco TelePresence System EX90 Videoconferencing Unit	\$ 22,638.48	-8.00%
GBC DocuBind P400 Electric Binding System	\$17,965.07	-10.45%
Lexmark X611dhe Monochrome Laser Printer	\$16,829.90	-27.27%
Martin Yale Chadless Opener Electric Letter Opener	\$16,656.20	-7.80%
Cubify CubeX 3D Printer Double Head Print	\$11,099.63	-80.00%
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	\$9,917.64	about -29.00%
Bush Advantage Collection Racetrack Conference Table	\$9,547.25	about -20.27%
Cubify CubeX 3D Printer Triple Head Print	\$7,999.98	-48.00%

At product level, the dashboard gives clearer evidence that some high-sales products are actually damaging profit. For example, Cisco TelePresence System EX90 has the highest sales among the listed products, around \$22,638.48, but its profit margin is still -8.00%. This means the product looks strong from a sales view, but it is not profitable.

Other examples show a similar issue. GBC DocuBind P400 has around \$17,965.07 sales but a -10.45% margin, while Lexmark X611dhe has around \$16,829.90 sales but a much weaker -27.27% margin. The most serious example is Cubify CubeX 3D Printer Double Head Print, which has around \$11,099.63 sales but a very poor -80.00% margin. This product is especially risky because most of its sales value is being lost through cost, discount, or pricing problems.

The key insight is that the business should not judge product performance by sales alone. A product can appear successful because it generates high revenue, but if its profit margin is negative, it reduces overall business performance. Products such as Cisco TelePresence System EX90, GBC DocuBind P400, Lexmark X611dhe, and Cubify CubeX 3D Printer should be reviewed carefully.

Overall, Technology should remain the main profit-driving category, while Furniture needs the most attention because it has the weakest net profit and the highest loss amount. The business should also investigate repeated loss areas such as Binders, Chairs, Tables, Furnishings, and Storage. For high-sales but negative-profit products, the business should review pricing, discount levels, supplier costs, shipping costs, and whether these products should continue to be promoted.

5.8 Temporal Trends

Business question: Based on order date, are there specific months, quarters, or seasons where the Superstore experiences significant peaks or dips in sales and profit?

Table 20: Quarterly Sales and Profit Seasonality

Period	Pattern	Interpretation
Q1	Lower sales and profit.	Weaker start of the year.
Q2	Gradual improvement.	Business activity begins to increase.
Q3	Sales and profit continue rising.	Stronger mid-to-late year performance.
Q4	Highest sales and profit in most views.	Main seasonal peak.

The temporal trend analysis shows that sales and profit are not evenly distributed across the year. The Superstore experiences clear seasonal changes, with weaker performance at the beginning of the year and stronger performance toward the end of the year.

Q1 is generally the weakest period, with lower sales and profit. This suggests that customer demand is slower at the start of the year, so the business may experience fewer orders or lower-value purchases during this period.

Q2 shows gradual improvement, meaning sales and profit begin to recover after the weaker Q1 period. This indicates that business activity starts to increase as the year progresses.

Q3 continues the upward trend, with stronger sales and profit compared with the first half of the year. This suggests that customer demand becomes more active during the mid-to-late year period.

Q4 is the strongest quarter in most views, showing the highest sales and profit. This indicates a clear year-end seasonal peak, possibly linked to holiday shopping, year-end purchasing, promotional campaigns, or business spending before the end of the year.

Across the full 2023–2026 period, sales and profit generally increased over time. The highest total sales and total profit occurred in 2026, showing that the business achieved revenue growth. However, the highest profit margin occurred in 2025, not 2026. This is an important insight because it shows that higher sales do not always mean better profit efficiency. In 2026, the business generated more revenue, but its margin was not as strong as the previous year.

Overall, the key insight is that Q4 is the main peak period, while Q1 is the weakest period. The Superstore should prepare inventory, marketing campaigns, staffing, and delivery capacity before Q4 to take advantage of the seasonal demand. However, during high-sales periods, the business should still monitor discounts, product mix, and profit margin carefully so that sales growth does not reduce profitability.

6. Recommendations

Table 21: Strategic Recommendations Based on Key Findings

Key issue identified	Deeper business insight	Recommended action	Expected benefit
Furniture has high sales but very weak net profit	Furniture is not necessarily a sales problem, but a margin problem . Tables and Chairs may sell well but are affected by high cost, discounting, or shipping expense.	Conduct a detailed margin review for Tables, Chairs, and Furnishings . Adjust prices, reduce unnecessary discounts, and review supplier or delivery costs.	Improves Furniture profitability without needing to reduce sales volume.
Tables and Chairs are repeated Furniture loss drivers	Losses are concentrated in specific sub-categories, so broad category-level decisions may be too general.	Create a sub-category action plan: review product-level margin, remove or reduce promotion on unprofitable items, and prioritise higher-margin Furniture products.	Reduces repeated losses from the weakest Furniture areas.
Binders have the highest number of loss-making orders	Binders may not create the largest loss per order, but the high number of loss orders means they create a repeated profitability leak.	Review Binder pricing rules, discount thresholds, and minimum order value. Limit discounts where the margin becomes negative.	Reduces frequent small losses that accumulate over time.
High-sales products can still have negative margins	Products such as Cisco TelePresence, GBC DocuBind, Lexmark printer, and Cubify 3D printers look successful by sales, but they damage profit.	Flag products with high sales + negative margin for management review. Decide whether to increase price, reduce discounts, renegotiate cost, or stop promoting them.	Prevents the business from chasing revenue that reduces overall profit.
Texas has high sales but weak profitability	Texas is a good example of why sales alone can be misleading. The issue may come from product mix, discounts, or shipping choices.	Compare Texas against profitable states such as California and New York. Identify whether losses are caused by specific products, high discounts, or shipping modes.	Turns a high-revenue but risky state into a more profitable market.
Low-profit cities appear repeatedly	Cities such as Houston, Philadelphia, Chicago, Dallas, Phoenix, and Aurora may be reducing regional profit.	Build a city-level profitability review using sales, profit margin, discount level, product category, and ship mode.	Helps identify local causes of poor profit and supports targeted fixes.
Medium and high discounts reduce profit	Discounts are helping generate orders, but medium and high discounts do not appear to increase sales enough to justify the profit loss.	Set discount control rules. Use low discounts for targeted promotions and require approval for medium or high discounts, especially for weak-margin products.	Protects profit margin while still allowing controlled sales growth.

Discount strategy is not connected enough to product profitability	Applying discounts broadly can make already weak products even less profitable.	Apply discount limits by product category and sub-category. For example, stricter discounts for Furniture, Tables, Chairs, Binders, and negative-margin products.	Reduces the chance of selling risky products at a loss.
Consumer dominates sales	Consumer is the main revenue driver, but relying too much on one segment may create dependence.	Maintain Consumer-focused marketing, but develop Corporate customers as a stable secondary segment through targeted offers and relationship-building.	Supports long-term growth and reduces dependence on one segment.
Home Office is the smallest segment	Home Office has lower total contribution, but some selected state/product combinations may still be profitable.	Do not remove focus completely. Instead, analyse Home Office by state and product margin before reducing investment.	Avoids losing small but potentially profitable customer groups.
Faster shipping does not guarantee higher profit	Same Day and First Class may improve service speed, but they can be less stable because of lower order volume and possible higher delivery cost.	Monitor profit per order by ship mode . Avoid using fast shipping for low-margin or heavily discounted orders unless justified.	Improves shipping efficiency and prevents delivery cost from reducing margin.
Standard Class appears more stable	Standard Class likely supports higher order volume and more consistent profit per order.	Promote Standard Class for regular orders and use faster shipping only for high-value or strategically important orders.	Balances customer service with profitability.
Q4 is the strongest seasonal period	Q4 brings the highest sales and profit, but it may also encourage heavier discounting and operational pressure.	Prepare inventory, staffing, and marketing before Q4, but set discount limits and monitor profit margin weekly during the peak period.	Captures seasonal demand without sacrificing profitability.
2026 has the highest sales/profit, but 2025 has the highest margin	Revenue is growing, but profit efficiency may not be improving at the same speed.	Track sales growth together with profit margin, not separately. Review whether 2026 growth came from discounting or low-margin products.	Ensures future growth is profitable, not just larger in sales value.

7. Limitations

This report has several limitations. First, the analysis is mainly descriptive, meaning it identifies patterns in sales, profit, discounts, regions, products, and shipping, but it cannot fully explain the causes behind those patterns. For example, the dashboard shows that Furniture has weak profitability, but it cannot prove whether this is caused by high discounts, supplier costs, shipping costs, or product mix.

Second, the findings show relationships rather than causation. High discounts are associated with lower profit, but the dashboard alone cannot prove that discounts are the only reason for the loss. Other factors such as product category, region, order size, or shipping mode may also affect profit.

Third, the report is limited by the available dataset. It does not include external business information such as customer satisfaction, competitor pricing, supplier contracts, market demand, or detailed operating costs. Therefore, the recommendations should be treated as data-supported suggestions rather than final business decisions.

Finally, some insights are based on aggregated dashboard views. While this helps identify overall trends, it may hide differences at product, city, or order level. Further analysis is needed before implementing major changes, especially for discount strategy, Furniture pricing, and regional profitability improvement.

8. Conclusion

This report analysed the Superstore's sales and profit performance across products, regions, customer segments, discounts, shipping modes, and time periods. Overall, the findings show that high sales do not always lead to strong profitability. Some categories, states, and products generate large sales volumes but still create weak or negative profit margins.

The strongest-performing category is Technology, which has the highest net profit and relatively lower losses. Office Supplies is also a strong contributor, but repeated losses from Binders reduce its profitability. In contrast, Furniture is the weakest category because it has the lowest net profit and the largest loss amount, especially from Tables, Chairs, and Furnishings.

The regional analysis shows that California and New York are the strongest markets because they generate high sales and generally positive profit margins. However, Texas and several low-profit cities show that high revenue does not always mean strong profit. These areas need further investigation into product mix, discounts, and shipping choices.

The discount analysis also highlights a major issue. No discount and low discount orders are generally healthier for profit, while medium and high discounts create a higher risk of negative margins. This suggests that the business should control discounting more carefully instead of using discounts broadly to increase sales.

Customer segment analysis shows that Consumer is the main revenue driver, while Corporate is an important secondary segment with stable long-term value. Home Office contributes less overall but may still be useful in selected states or product groups. The shipping analysis suggests that faster shipping does not automatically improve profit, so ship mode should be evaluated together with profit per order and order volume.

In terms of time trends, the Superstore performs strongest in Q4, while Q1 is generally weaker. Sales and profit increased from 2023 to 2026, but the highest profit margin occurred in 2025. This means the business is growing, but profit efficiency must still be monitored.

Overall, the key conclusion is that the Superstore should move from focusing only on sales growth to focusing on profit-quality growth. The business should prioritise profitable categories, review weak-margin products and regions, control high discounts, and prepare carefully for Q4 demand while protecting profit margins. This approach can help the Superstore improve long-term profitability rather than only increasing revenue.

Appendix

DAX Measures

Measure	DAX / Definition	Purpose
Total Sales	SUM(FactSales[Sales])	Total revenue generated.
Total Profit	SUM(FactSales[Profit])	Total profitability.
Total Quantity	SUM(FactSales[Quantity])	Total units sold.
Total Orders	DISTINCTCOUNT(FactSales[Order ID])	Number of orders.
Total Customers	DISTINCTCOUNT(FactSales[Customer ID])	Number of customers.
Profit Margin %	DIVIDE([Total Profit], [Total Sales])	Profitability efficiency.
Average Discount	AVERAGE(FactSales[Discount])	Average discount rate.
Profit per Order	DIVIDE([Total Profit], [Total Orders])	Average profit generated per order.
Average Shipping Days	AVERAGE(FactSales[Shipping Days])	Average delivery duration.
Loss Orders	CALCULATE(DISTINCTCOUNT(FactSales[Order ID]), FactSales[Profit] < 0)	Number of loss-making orders.
Loss Order %	DIVIDE([Loss Orders], [Total Orders])	Proportion of orders that generated losses.
Loss Amount	CALCULATE([Total Profit], FactSales[Profit] < 0)	Total negative profit amount.
Profit Amount	CALCULATE([Total Profit], FactSales[Profit] > 0)	Total positive profit amount.

Visual Construction Guide

Dashboard section	Visual to create	Power BI fields / settings
Executive Overview	Cards	Add card visuals for Total Sales, Total Profit, Profit Margin %, Total Orders, and Total Customers.
Executive Overview	Sales & Profit Trend	Line chart: Quarter Name on X-axis; Total Sales and Total Profit as values. Sort Quarter Name by Quarter.
Executive Overview	Category comparison	Clustered column chart: Category on X-axis; Total Sales and Total Profit as values.
Product Profitability	Profit/Loss toggle	Use Profit Status slicer or buttons/bookmarks to switch between Loss and Profit views.

Product Profitability	Top 10 Lowest Profit Products	Table: Product Name, Total Sales, Total Profit, Profit Margin %. Sort by Total Profit ascending.
Product Profitability	Top 10 Most Profitable Products	Table: Product Name, Category, Sub-Category, Total Sales, Total Profit, Profit Margin %. Apply Top N = 10 by Total Profit descending.
Regional & Customer Segment	Top states visual	Combo chart: State on X-axis; Total Sales as columns; Profit Margin % as line.
Regional & Customer Segment	Segment sales by state	Stacked column chart: State on X-axis; Segment as legend; Total Sales as value.
Discount, Shipping & Trend	Discount impact table	Table: Discount Level, Total Sales, Total Quantity, Total Profit, Profit Margin %, Loss Order %. Sort by Discount Sort.
Discount, Shipping & Trend	Shipping efficiency visual	Scatter chart: Average Shipping Days on X-axis; Profit per Order on Y-axis; Total Orders as size; Ship Mode as slicer.